Exercice 6 :

$$g(w) = \frac{w^T C w}{w^T w}$$

We have :

$$\nabla g(w) = \frac{2Cw \cdot (w^T w) - w^T C w \cdot (2w)}{(w^T w)^2}$$

$$= \frac{2}{w^T w} \left(Cw - \frac{w^T C w}{w^T w} w \right)$$

$$\nabla g(w) = 0 \iff Cw = \left(\frac{w^T C w}{w^T w} \right) w$$

then the stationary points w of $g(w)$ are the eigenvectors of C corresponding to the eigenvalues $\frac{w^T C w}{w^T w}$ the Rayleigh quotient.